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Q17.132

$$\begin{cases} mg \cos \beta - N = m \left(\frac{L}{2} \right) \omega^2 & \text{--- ①} \end{cases}$$

$$\begin{cases} m g \left(\frac{L}{2} \right) (1 - \cos \beta) = \frac{1}{2} I_B \omega^2 & \text{--- ②} \\ = \frac{1}{2} \left(\frac{1}{3} m L^2 \right) \omega^2 \end{cases}$$

$$\begin{cases} N = 0, \therefore \text{loss contact} \end{cases}$$

$$\begin{cases} I_B = \frac{1}{3} m L^2 \end{cases}$$

$$\begin{aligned} \text{①} \quad & \cos \beta = \frac{L \omega^2}{2g} \\ \text{②} \quad & 1 - \cos \beta = \frac{L \omega^2}{3g} \end{aligned}$$

$$\text{①} + \text{②} \quad \frac{5 L \omega^2}{6} = 1$$

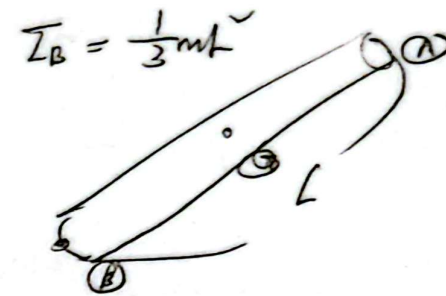
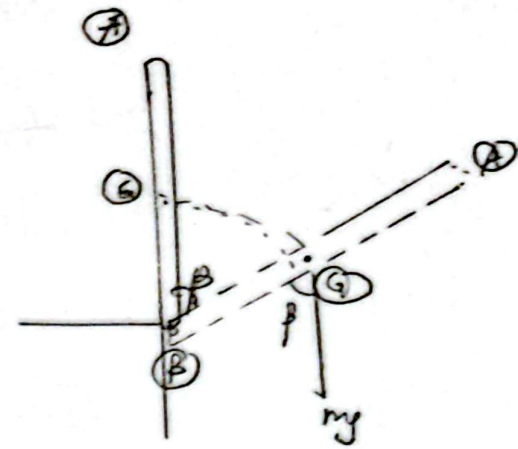
$$\therefore \beta = \arccos\left(\frac{3}{5}\right) \approx 53.13^\circ$$

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$$\text{c) } \therefore \omega = \sqrt{\frac{6g}{5L}}$$

$$v_A = \omega L = \sqrt{\frac{6}{5} g L} \approx 1.095 \sqrt{g L}$$

$\angle 53.13^\circ$



Q. 17.144 (a) $\rightarrow m_B = 18 \text{ lb}$

Same material $\rho = \text{const.}$

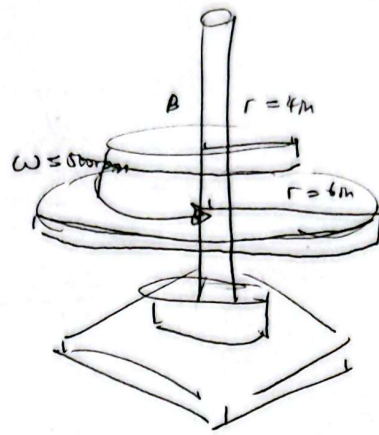
$$\rho = \frac{m}{A \cdot k}$$

$$m \propto A \propto r^2$$

$$\frac{m_B}{m_A} = \left(\frac{r_B}{r_A}\right)^2$$

$$m_B = m_A \left(\frac{r_B}{r_A}\right)^2$$

$$= (18) \left(\frac{4}{6}\right)^2 = 8 \text{ lb}$$



$$\begin{cases} I_B = \frac{1}{2} m_B r_B^2 = 2.25 \text{ lb ft}^2 \\ I_A = \frac{1}{2} m_A r_A^2 = 0.44 \text{ lb ft}^2 \end{cases}$$

$\rightarrow$ Conservation of Angular Momentum

$$I_A \omega_0 = (I_A + I_B) \omega$$

$$n = \frac{I_A}{I_A + I_B} n_0 \approx 417.5 \text{ rpm}$$

$$\vec{L} = \vec{r} \times m\vec{v} \Leftrightarrow I\vec{\omega}$$

$$\vec{L} = L_0$$

(b) $\Delta E = E_2 - E_1$

$$= \frac{1}{2} I_A \omega_0^2 - \frac{1}{2} (I_A + I_B) \omega^2$$

$$\omega = n \left(\frac{2\pi}{60} \right)$$

$$= \frac{1}{2} \left(\frac{2\pi}{60} \right)^2 \left[(2.25) - (2.25 + 0.44) \right]$$

$$\Delta E \approx \text{513 (unit of energy)}$$

$$\text{lb ft}^2/\text{s}^2$$

Q17.146 (a) (Conservation of A.M)

$$\therefore L_0 = L'$$

$$I_A \omega_0 - \vec{r}_{AG} \times m \vec{v} = -I_A \omega$$

$$- \left(\frac{2}{5} m r^2 + m_s d^2 \right) \omega$$

$$I_A = \frac{1}{3} m L^2, \quad d^2 = r^2 + L^2$$

$$\therefore \omega = 1.286 \text{ rad/s} \quad (\text{CW})$$

(b)

$$\therefore \sum M_A = I_A \alpha$$

$$\sum M_A = \sum \frac{dL}{dt} = \frac{d(I\omega)}{dt} = I_A \alpha$$

$$m_r g \left(\frac{L}{2} \right) + m_s g L = (I_A + (I_G + m_s d^2)) \alpha$$

$$m_r g \left(\frac{L}{2} \right) + m_s g L = \left[\frac{1}{3} m_r L^2 + \frac{2}{5} m_s r^2 + m_s (r^2 + L^2) \right] \alpha$$

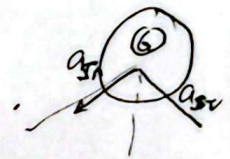
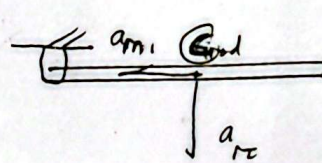
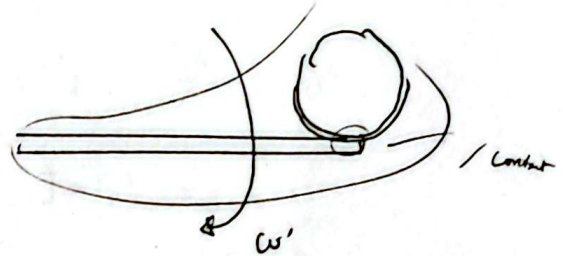
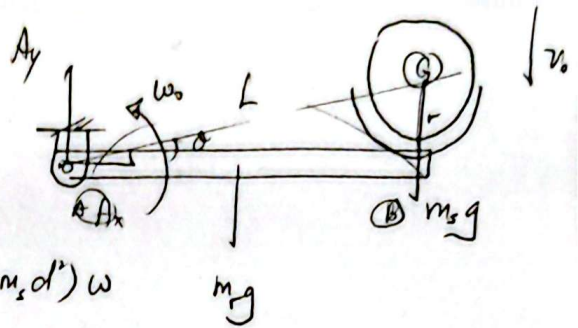
$$\alpha = 38.21 \text{ rad/s}^2$$

$$\sum F_y = \sum m a_y$$

$$\sum F_x = \sum m a_x$$

$$\left\{ \begin{array}{l} m_r g + m_s g - N_y = m_r a_{ry} \\ \quad \quad \quad + m_s (a_{sy} \sin \theta + a_{sx} \cos \theta) \\ N_x = m_s (a_{sx} \sin \theta - a_{sy} \cos \theta) - m_r a_{rx} \end{array} \right.$$

$$\left\{ \begin{array}{l} N_x = 0.719 \text{ lb} \rightarrow \\ N_y = 1.006 \text{ lb} \uparrow \end{array} \right.$$



$$\left\{ \begin{array}{l} a_{rn} = \omega^2 \left(\frac{L}{2} \right) \\ a_{rz} = \alpha \left(\frac{L}{2} \right) \end{array} \right.$$

$$\left\{ \begin{array}{l} a_{so} = \omega^2 \sqrt{r^2 + L^2} \\ a_{sz} = \alpha \sqrt{r^2 + L^2} \end{array} \right.$$